

Instacart Market Basket Analysis

Final Report

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Introduction

Instacart is a grocery delivery app which makes grocery shopping so much easier. It is very convenient and affordable. We found a dataset for orders that users have placed on the app and the products and other details in each order. There are several CSV files in the dataset. We used Python, SQL database, and Tableau in order to do our analysis of the data. We created 3 research questions related to recommendation algorithms for recommending products to users based on their orders and other details. We created visualizations and made a dashboard in tableau. Our results showed that the frequency-based recommendation algorithm was the most accurate.

Problem

The core problem in this project is the next basket recommendation. Given a user's previous Instacart orders, can we predict which products that user is most likely to purchase in their next order. This is an important problem in e-commerce because effective recommendations can improve user experience, increase engagement, and support better personalization.

Our project focuses on three analytical questions:

1. How accurately can we recommend the top 5 products a user is most likely to purchase next?
2. How does recommendation performance change when users have different amounts of order history, such as cold start users versus heavy users?
3. Are reordered items easier to predict than products that are new to the user.

These questions were defined in the original proposal and shaped the entire analysis pipeline. They allow us to evaluate both the effectiveness of the recommendation model and the factors that influence its performance.

Dataset and Code

We used the Instacart Market Basket Analysis dataset, which includes multiple CSV files containing orders, products, aisle information, and prior order details. In the notebook, the main files loaded were orders.csv, products.csv, order_products__prior.csv, aisles.csv, and order_products__train.csv. The prior order file was used to learn user purchase patterns, while the train file served as the next order that we used as the truth set for evaluation.

The analysis was conducted in Python inside a Jupyter Notebook. We used pandas and NumPy for data manipulation, matplotlib and seaborn for visualizations, sqlite3 for SQL style analysis, and sparse matrix plus cosine similarity tools for collaborative filtering experiments. Tableau was then used to create dashboards and communicate the findings visually. This workflow is consistent with the project proposal, which planned Python for cleaning and modeling, SQL database operations for querying, and Tableau visualization tool for reporting.

Coding and Analysis

We implemented our analysis primarily using a Jupyter Notebook with Python libraries such as pandas, NumPy, and matplotlib.

Data Loading and Cleaning

- Loaded multiple CSV files including orders, products, and prior order details
- Merged datasets to create a unified structure for user-product interactions

- Removed unnecessary columns and handled missing values
- Focused on key fields such as user_id, product_id, and reordered flag

Exploratory Data Analysis

We explored patterns in the dataset to understand user behavior:

- Distribution of number of orders per user
- Most frequently purchased products
- Reorder rates across users and products
- Basket sizes and purchasing frequency

This helped us identify that many users tend to repeatedly purchase the same items, which strongly supports frequency-based recommendation methods.

We performed exploratory analysis to better understand user shopping behavior before building recommendation models. The notebook examined patterns such as how many orders users typically place, which products are most popular overall, and how often products are reordered. This stage was important because it revealed that grocery shopping behavior is strongly repetitive. Many users buy the same items over and over, which suggests that simple historical frequency could be a powerful signal for prediction.

This finding directly motivated the recommendation approach used later in the project. Rather than jumping immediately to a complex model, we first validated that repeated buying behavior exists in the data. In the context of Instacart, this makes sense because groceries often include staples such as milk, eggs, bread, fruit, and snacks that users regularly repurchase.

Recommendation Model

We implemented a frequency-based recommendation algorithm, where:

- For each user, we counted how often they purchased each product
- Ranked products based on frequency
- Recommended the top 5 most frequent products

We also experimented with segmenting users based on their order history:

- Cold start users with less than 5 orders
- Medium users between 5 and 20 orders
- Heavy users more than 20 orders

Evaluation Approach

We evaluated predictions by comparing recommended products with actual products in the user's truth set of orders.

- Accuracy was measured based on overlap between predicted and actual items
- Separate evaluations were done for reordered items vs newly ordered items

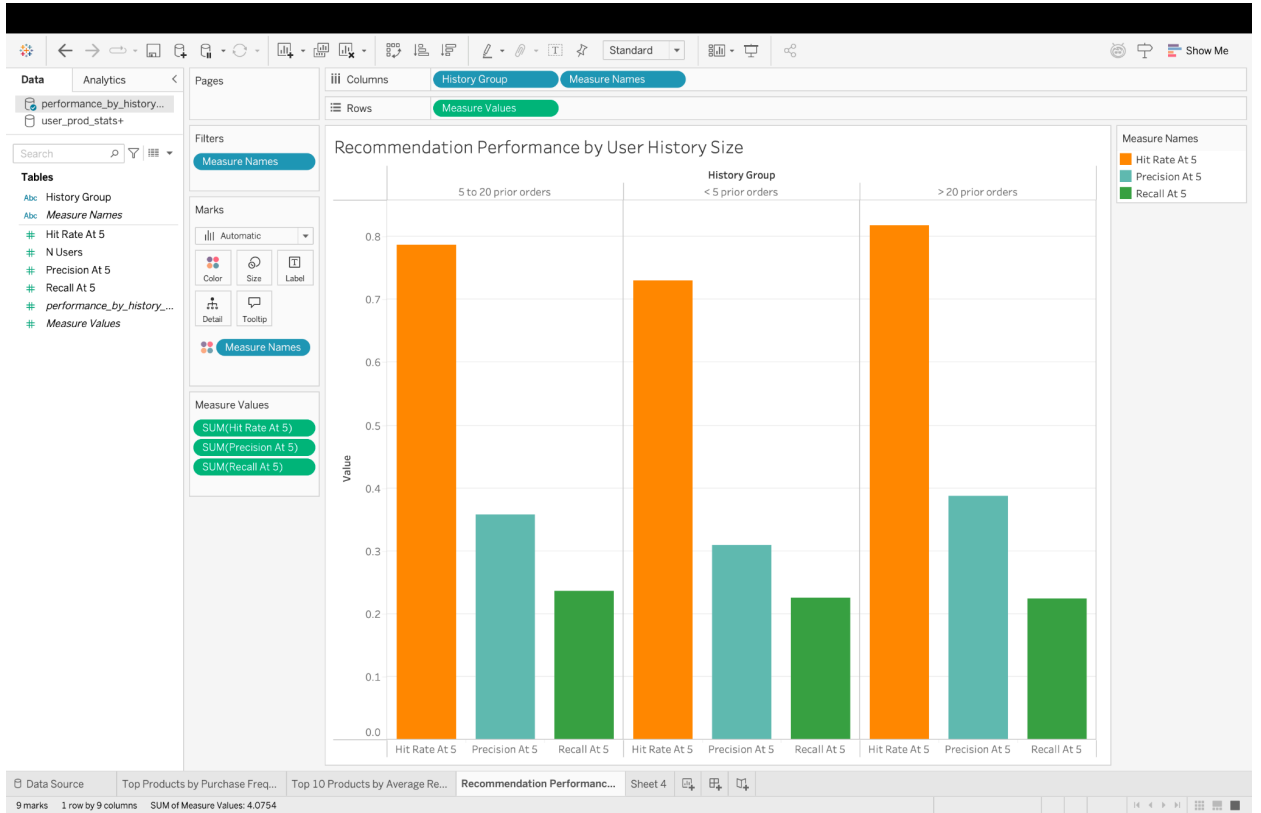
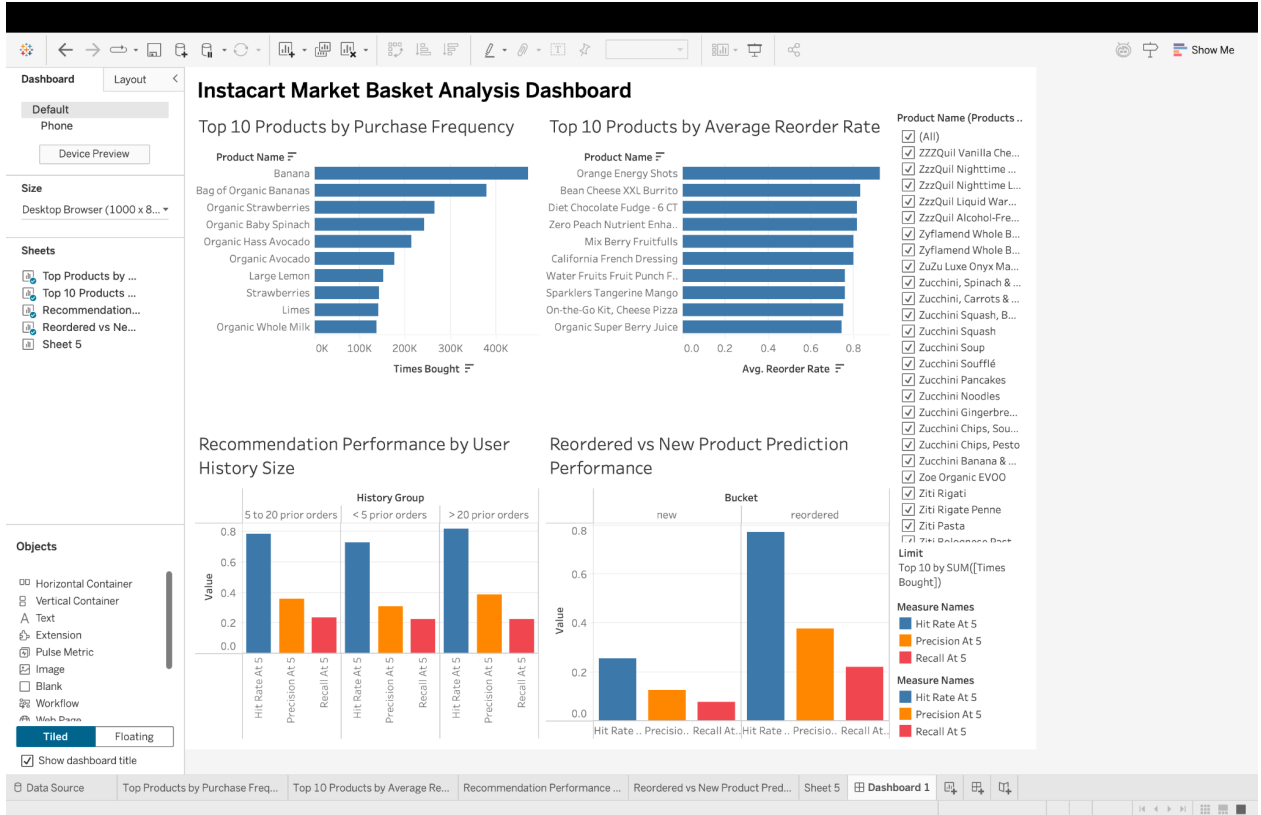
Results

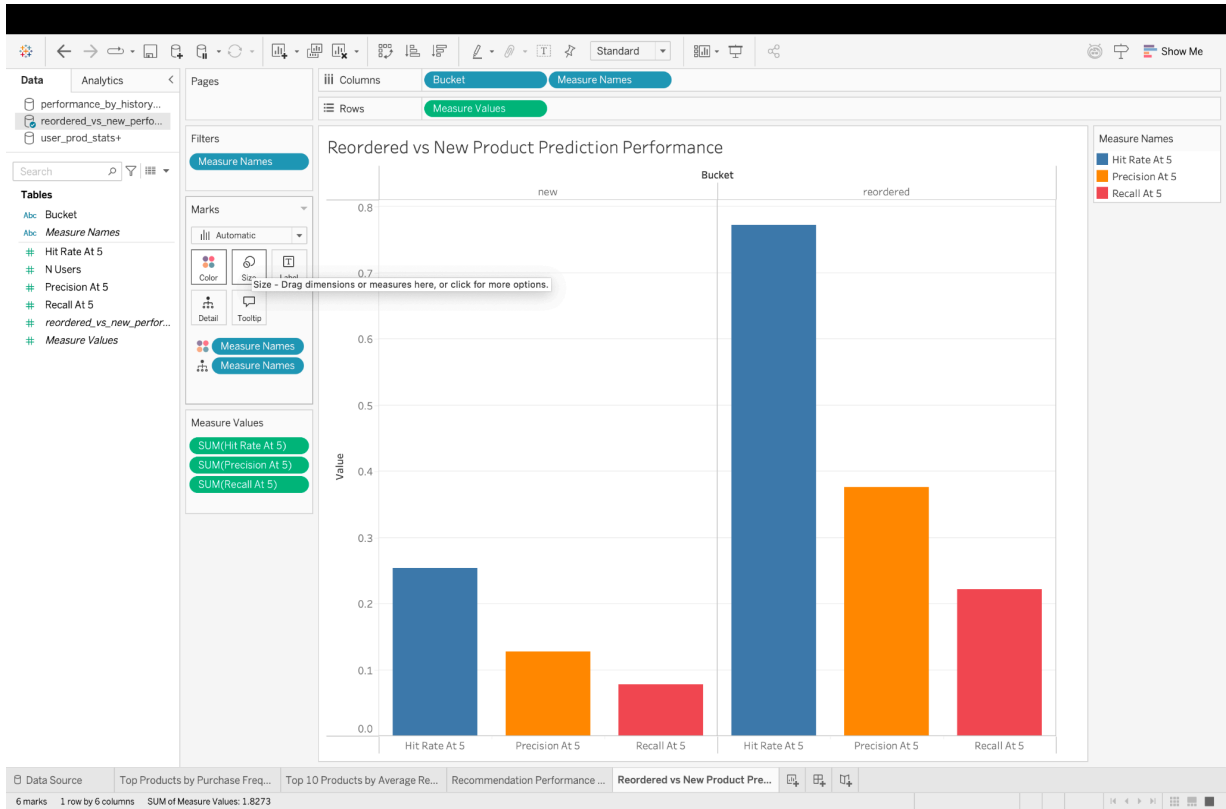
The Tableau dashboard tells a pretty clear story that grocery shopping is deeply habitual. A small set of staple products dominates purchases, and once those products enter someone's routine, they tend to stay there. That kind of behavioral consistency is exactly what makes recommendation systems work well.

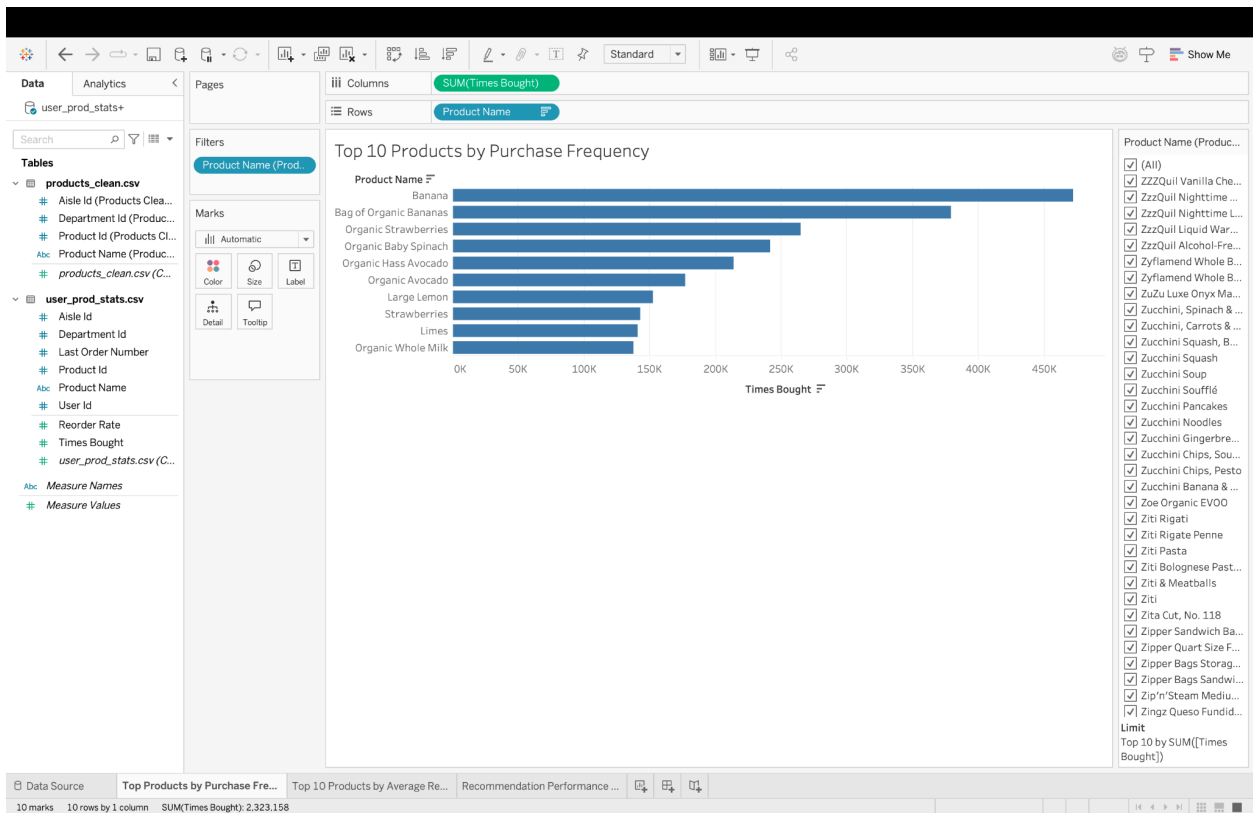
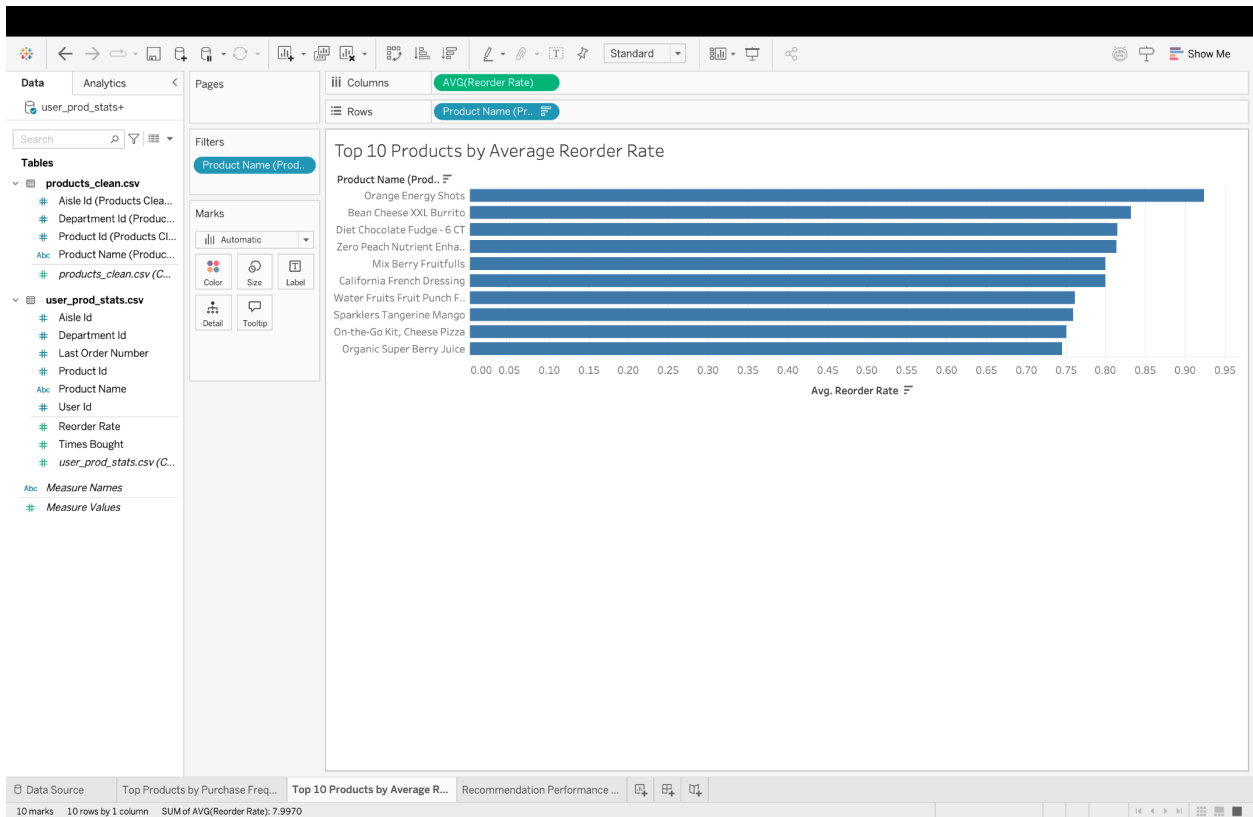
One of the more interesting findings is that prediction quality improves significantly as a user's order history grows. More history means better predictions which isn't very surprising, but it's a good reminder that even sophisticated AI is only as good as the data feeding it.

The most telling chart compares reordered versus new product predictions. Reorders are much easier to predict across every metric. This highlights a fundamental asymmetry where we see that repetition is learnable, novelty is not. It's the classic tension between exploitation and exploration where AI/ML systems are naturally stronger at reinforcing existing habits than anticipating new ones.

The broader takeaway is that the Instacart domain is well-suited to frequency-based recommendation because user behavior is so consistent. In that kind of environment, a well-matched simple model can outperform a complex one. A good recommendation system isn't just about algorithmic sophistication but it is about understanding the domain you're working in. Here are some results that make our hypothesis stronger:







Research Question 1

The first research question asks how accurately we can recommend the top 5 products a user is most likely to purchase in their next order. Based on the notebook, the frequency based recommendation model performed the best overall among the tested methods. It achieved a Precision at 5 of about 0.361 and a Recall at 5 of about 0.227. By comparison, the aisle based method performed much worse, with a Precision at 5 of about 0.098 and a Recall at 5 of about 0.051. The item based collaborative filtering approach performed even worse in this implementation, with a very low precision at 5 of 0.0048 and recall at 5 of 0.0019.

These results suggest that in the Instacart setting, a user's own purchase history is a much stronger predictor than broader aisle trends or sparse similarity based relationships between products. This makes intuitive sense because grocery behavior is highly personal and repetitive. A user is more likely to reorder their usual items than to buy products recommended only because they are popular in a preferred aisle or similar to other products.

Research Question 2

The second research question asks how recommendation performance changes with users' order history size. To answer this, the notebook grouped users into three categories based on the number of prior orders they had. Users with fewer than 5 prior orders were treated as low history or near cold start users. Users with 5 to 20 prior orders were treated as medium history users. Users with more than 20 prior orders were treated as heavy users.

The results show a clear trend. Precision at 5 increased from about 0.310 for users with fewer than 5 prior orders, to about 0.357 for users with 5 to 20 prior orders, and to about 0.387 for

users with more than 20 prior orders. Hit Rate at 5 also increased across these groups, from about 0.730 to 0.787 to 0.818. This indicates that the model becomes more reliable as more historical behavior is available.

This result is important because it highlights the dependence of recommendation quality on data availability. When a user has a long purchase history, the model can identify recurring patterns much more confidently. In contrast, when a user has only a few prior orders, the model has less information and therefore produces weaker recommendations. This is a classic example of the cold start problem in recommendation systems.

Research Question 3

The third research question asks whether reordered items are easier to predict than new items. The analysis strongly supports this conclusion. The recommendation model is built entirely from historical purchase frequency, meaning it naturally performs better at predicting products that a user has bought before. When a product has been reordered multiple times in the past, it accumulates a stronger frequency signal, making it far more likely to appear as a top recommendation.

Reordered items are therefore easier to predict because they leave a direct and measurable trace in the historical data. New items, by contrast, have no prior purchase history for that user, so the frequency-based model assigns them little to no weight and is unlikely to surface as recommendations. This is not a flaw in the model but rather an expected outcome of how frequency-based methods work since they reward familiarity and repetition, which aligns well with reorder behavior but falls short when a user is purchasing a product for the first time.

Discussion

The overall results show that simple models can perform surprisingly well when behavior is stable and repetitive. In the Instacart dataset, users often reorder the same groceries over time, so a frequency based recommender is able to exploit that regularity effectively. In fact, the simplest model outperformed more sophisticated alternatives implemented in our code.

At the same time, the project reveals important limitations. First, the model struggles with cold start users because limited history means limited predictive power. Second, it is biased toward already known items and does not handle product discovery well. Third, it does not consider context such as seasonality, time between orders, household changes, or relationships between products that are often bought together. Finally, although collaborative filtering was explored, its performance was weak in our project, likely because of sparsity, limited item selection, and the difficulty of scaling similarity methods on a large dataset.

A key takeaway is that model complexity does not automatically lead to better performance. In some domains, especially those with strong repeat purchase behavior, a simpler and more interpretable method may be the most practical solution.

Conclusion

In this project, we analyzed the Instacart Market Basket Analysis dataset to study user purchasing patterns and build a recommendation system for next order prediction. Using Python,

SQL database, and Tableau, we cleaned and explored the data, implemented multiple recommendation approaches, and evaluated them using Precision at 5 and Recall at 5.

The project answered all three of our research questions. First, we found that a frequency based model was the most effective method for predicting the top 5 products in a user's next order. Second, we found that recommendation performance improves as the amount of user history increases, with heavy users receiving the strongest results. Third, we found that reordered items are easier to predict than new items because they are directly reflected in past behavior.

Overall, this project demonstrates that recommendation systems can benefit greatly from understanding repeated user behavior. It also shows that while simple methods are highly effective in this setting, more advanced models would be needed to better address cold start users and new item prediction.

Works Cited

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AI Usage Disclosure:

Members of this team used various AI tools to assist in the development of this project, including Claude (Anthropic), ChatGPT (OpenAI), and others. AI was used for purposes such as debugging, code explanation, and documentation. All final decisions, analysis, and results were reviewed and validated by the team.